

The Content-Dependent Limits of Acoustic Compression

Learning what acoustic evidence to retain—and where to compress it—under a fixed memory budget

Proposal for discussion with Professor Richard Zemel

Central question. Can a speech-native omni model use a fixed memory budget more effectively by preserving acoustic representations according to their *marginal value*—the information they add beyond text and other retained audio—and by compressing them only after the representation contains enough context to identify that value?

Working hypothesis. Acoustic history is not uniformly compressible. When a transcript already carries the relevant content, additional acoustic positions can be redundant; when text is incomplete, conflicting, or insufficient for prosody, speaker, or background sound, the same positions can be essential. Compression is therefore bounded by the evidence the transcript and retained state fail to carry, and that boundary depends jointly on content, audio reliability, and representation depth.

From VoxZip to content-dependent acoustic memory

VoxZip is the immediate prior work and starting mechanism. It uses ASR timestamps to divide audio into semantic intervals, mean-pools each interval to the number of transcript tokens, and adds each pooled acoustic vector to its matching text embedding. It then reduces the growing KV cache using temporally decayed attention [1]. VoxZip therefore answers an important first question: transcript-anchored pooling can preserve useful audio while reducing long-context cost.

Its compression rule is fixed by the transcript. An interval with m_i transcript tokens receives m_i pooled acoustic positions, and fusion is element-wise addition. That leaves open whether every interval deserves the same acoustic budget relative to its transcript, whether a learned fusion can preserve information that addition loses, and whether compression should occur before the Thinker or after early Thinker layers have integrated acoustic and linguistic evidence. SpeechKV provides the strongest contrasting placement: it passes full speech through early language-model layers and learns to pool keys and values while retaining full-resolution queries and residual states [2].

We express the extension as two separate decisions:

$$\underbrace{b_i}_{\text{positions retained for segment } i} = g_\phi(z_i, B), \quad \sum_i b_i \leq B, \quad \underbrace{H_i}_{\text{memory passed onward}} = F_\theta(T_i, P_{b_i}(A_i)).$$

Here A_i is the acoustic sequence, T_i its transcript representation, z_i the evidence available to the allocator, P_{b_i} a compressor that returns b_i acoustic positions, F_θ the fusion rule, and B the conversation-level budget. VoxZip uses transcript length to set b_i and fixed addition for F . We propose to learn how much acoustic memory each segment receives, what its retained representation contains, and where that representation can be compressed safely.

Preliminary result: acoustic value is heterogeneous

CleaRx ran three frozen-Qwen3-Omni studies that progressively separated fusion strength from actual memory allocation [3]. A one-clip smoke test showed that one transcript position plus mean-pooled audio moved the model closer to its full-audio continuation: teacher KL fell from 2.136 to 1.966. Across ten clips, fixed-strength pooled audio improved mean KL by 7% over transcript-only and improved 7/10 items. This established that aggressively pooled audio could still carry useful evidence into the frozen Thinker.

The next study learned a continuous audio–text weight. It recovered the intended ordering of exact, incomplete, and conflicting transcript conditions, but its held-out KL was 5.8% worse than the best fixed weight. The result redirected the work: scaling a pooled vector changes its influence but does not save positions. The relevant decision is how to spend a fixed number of acoustic positions.

The crossed-retention pilot held that budget at 48 positions. Uniform retention kept two positions for every one of 24 test items; concentrated policies kept four positions for 12 items and zero for 12. Agreement-based allocation reduced mean KL by 7.96% relative to uniform retention. It did not beat the audio-only allocator, and the improvement over uniform retention changed sign across the two held-out speakers. The allocator is therefore not the result. The result is the measured structure of acoustic value:

Benefit of four acoustic positions over transcript-only	Clean audio	Degraded audio
Exact transcript	0.0025	0.0011
Incomplete transcript	4.5444	1.7199
Conflicting transcript	4.3288	0.8755

What the result changes

In this controlled fixture, acoustic positions added almost nothing when the transcript was exact, added substantially more when text omitted or contradicted usable speech, and lost value when the audio was degraded. This is direct evidence that equal-length segments can have sharply different marginal acoustic value. It motivates the research problem without claiming that a linear heuristic trained on independent clips has solved conversation-level allocation.

Aim 1: learn marginal acoustic value under a shared budget

Construct short multi-turn histories with natural and controlled transcript errors, clean and degraded audio, and lexical plus paralinguistic queries. Measure the benefit curve

$$u_i(k) = \mathcal{L}(T_i) - \mathcal{L}(T_i, P_k(A_i)), \quad k \in \{0, 1, 2, 4, 8\},$$

where $\mathcal{L}$ combines full-sequence teacher divergence with task loss. Then learn g_ϕ to allocate B across all segments in a history. Compare uniform allocation, random concentrated allocation, audio-only confidence, the existing linear agreement score, a nonlinear local predictor, and an oracle computed from measured $u_i(k)$. This tests the conceptual step that the preliminary fixture could not: whether independently observed content effects support joint allocation under one conversation-level constraint.

Separately compare fixed mean pooling, method-faithful VoxZip pooling and addition, learned monotonic summarization, and learned fusion F_θ . This factorial design answers a second open question: whether a gain comes from *where the budget goes* or *what retained positions contain*. Use held-out speakers and conversations, paired examples, and clustered uncertainty. Continue with adaptive allocation only if it beats uniform, random, and audio-only policies at equal KV bytes; retain learned fusion only if it improves on VoxZip-style pooling and addition.

Aim 2: locate the earliest safe compression boundary

Compare the strongest Aim 1 pre-Thinker method with SpeechKV-style pooling after selected early and middle Thinker layers. First profile adjacent-state similarity, effective rank, and matched-pooling damage across Audio Tower outputs and Thinker layers. Then train only the small compressor or KV gate while freezing Qwen3-Omni. Match methods on retained KV bytes, not merely position count, and report compute spent before compression.

Primary outcomes are full-sequence teacher KL, lexical and paralinguistic task agreement, retained positions, peak KV bytes, prefill time, and decode time. Full audio is the quality reference; transcript-only is the zero-acoustic baseline. If in-Thinker pooling wins at the same memory budget, move the method inside the Thinker. If no compressed method preserves paralinguistic performance, increase the acoustic budget or narrow the claim to linguistic history. If fixed pooling matches learned methods, prefer the fixed method.

Expected contribution

The study will produce a controlled map of *what* acoustic information merits retention, *how much* memory it receives, and *where* it can first be compressed safely in Qwen3-Omni. The main contribution is not another global compression ratio. It is a content-dependent account of the boundary between redundant audio and acoustic evidence that text cannot replace, together with matched experiments that assign any improvement to allocation, representation, or compression location.

References

- [1] Liu et al. “VoxZip: Semantic-Anchored Temporal KV Cache Compression for Long-Context Audio Inference.” ACM MM, 2026.
- [2] Kim et al. “Compress the Cache, Not the Speech Embedding: KV Compression for Efficient Speech LLMs.” 2026.
- [3] CleaRx. [Exploratory Qwen3-Omni audio-memory experiments and retained results](#). September 2026.